INTERACTIVE DEVELOPMENT 300 · SEMESTER 2 · THEME 3

Trail.

Leave the worry at the trailhead.

Victor du Preez
Student No. 241250
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A THE STARTING HAND

The hand I was dealt.

Three cards, drawn at random in week 1. Rather than fight them, Trail makes all three pull in the same direction.

01 · GOAL

GOAL THEME

Reduce Stress

The lens for every feature, protect the calm that hiking already offers.

02 · DEVICE

DEVICE INTERACTION

Geolocation

The engine, live location, trail discovery and shared peace of mind.

03 · CONSTRAINT

CONSTRAINT

No Traditional UI Buttons

Gesture- and map-first; only the clear CTAs that safety truly needs.

B PROBLEM

Nature switches you off.

Worry keeps you switched on.

Time on the trail is one of the best ways to lower stress - but a hum of small worries chips away at the very reset you came for:

“ **Am I still on the path?**

“ **Have I had enough water?**

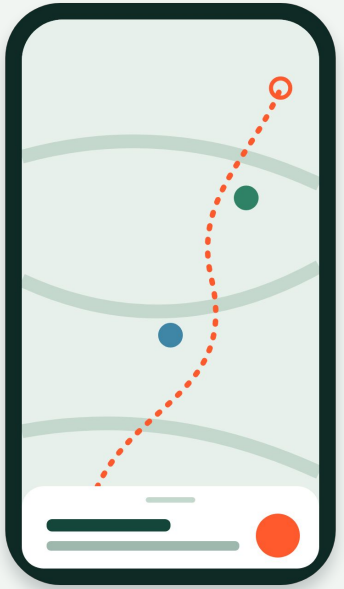
“ **Is everyone in my group okay?**

“ **Is my family worrying about me?**

Trail’s job: quietly absorb that load, so the hike can do what hiking is for.

A calm companion for stress-free hiking.

A cross-platform mobile app that helps hikers **discover trails, share their live location with the people who matter, and stay looked-after on the mountain** - through a quiet, gesture-first interface. It reframes a “safety” app as a calm one.



B WHO IT'S FOR

Four people, one worry-free hike.

H

Hiker

Standard user - discovers, saves, hikes and shares.

G

Group Organiser

Starts a shared hike and invites the party.

W

Watcher

Read-only family member who sees a live location from home.

T

The Trail Itself

Its plants, birds & safety notes - future guidance, on location.

C CORE FEATURES & SCOPE

Ship the calm first. Grow from there.

MVP MUST-HAVE

- Auth & gentle onboarding
- Trail search & discovery
- Save trails - CRUD data entity
- Live group location - Geo core
- Calm dashboard - trail · party · water

WK 4 Search, save, edit & remove a trail.

Nice-to-have PHASE 2

- Hydration & rest reminders
- Safety tips & survival snippets
- Share-location link / check-in

WK 5 Party positions update on a shared map within ~15s (2+ users).

Future LATER

- Plant & bird photo ID (ML)
- Offline trail maps
- Full survival library

WK 6 Hydration reminders fire as local notifications, on schedule.

One hike, start to finish.

STEP 01

01

Discover

Swipe through trails; save the ones that call.

STEP 02

02

Start hike

Pick a trail and begin - location wakes up.

STEP 03

03

Invite party

Share a link; friends & family join the map.

STEP 04

04

Stay together

Live map + water nudges keep everyone in view.

STEP 05

05

Peace of mind

Hike ends; sharing stops; everyone rests easy.

Built on the stack the brief requires.

Frontend React Native + Expo, in TypeScript

Gestures gesture-handler + reanimated

Backend Firebase - Auth - Firestore

Maps react-native-maps - trail & live views

Notifications expo-notifications - reminders

Live channel Realtime DB - throttled location writes

1 → 2 One library earns two requirements: `reanimated` powers the **gesture-first constraint** and the brief's required **interactive tweening / micro-interactions.**

Collections, schemas & data types.

users/{uid}

displayName	string
email	string
currentTrailId	string?
hydrationGoalMl	number

trails/{trailId}

name	string
difficulty	string
distanceKm	number
location	geopoint

.../savedTrails/{savedTrailId}

savedAt	timestamp
notes	string

groups/{groupId}

trailId	string
ownerId	string
memberIds	array
active	boolean

.../locations/{locationId}

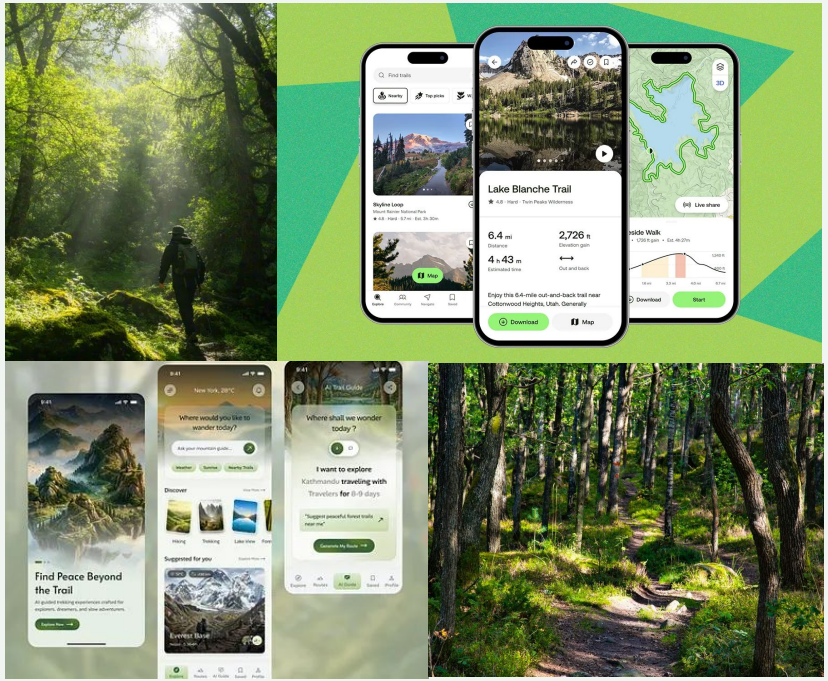
position	geopoint
updatedAt	timestamp

Relationships users 1—N savedTrails · users N—N groups · groups 1—1 trails · groups 1—N locations

A quiet, gesture-first feel.

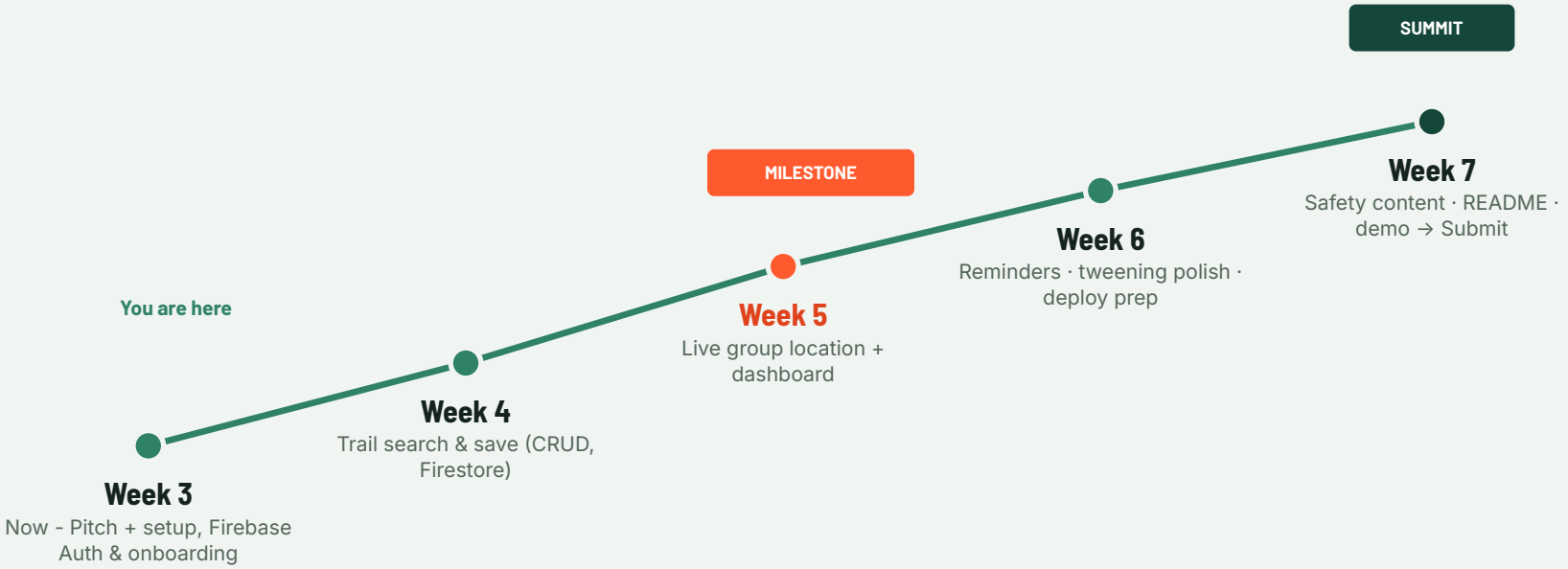
- **Swipe** cards - up saves, down skips
- **Pan / pinch** the map to share
- **Drag-up** sheet for detail
- **Pull** to refresh positions
- **Pager** not a tab bar
- Clear CTAs where safety matters

Accessibility: one-handed thumb-zone layout, high-contrast for sunlight, large tap targets, haptic confirmation for gestures, and a reduced-motion option.




Home / Map
Trail Detail
Discover
Group live

Week 3 to the summit - a five-week sprint.



G · IN CONCLUSION

A safety app, reframed as a calm one.

Three random cards - one coherent idea. Honestly scoped, and fully buildable on React Native + Firebase.

Trail.